

Dr. Divyesh Mistry

PhD, IIT Bombay, Mumbai, India-400076

✉ d.mistryg@gmail.com ☎ +1 (617) 396-0423
🌐 divyesh-mistry.github.io/webpage/ 🔗 linkedin.com/in/dr-divyesh-mistry-7b40a95b/
🐙 github.com/divyesh-mistry 📄 Google Scholar

Professional Summary

Computational Materials Scientist with over five years of experience developing atomistic and multiscale models of deformation, defect evolution, and mechanical behavior in advanced materials. Specializes in large-scale molecular dynamics simulations using **LAMMPS** and high-performance computing, complemented by discrete dislocation dynamics and continuum modeling. Develops automated **Python** workflows for simulation management, trajectory processing, defect characterization, and large-scale data analysis. Research spans tungsten alloys, nickel-based superalloys, and polymeric adhesive systems, with relevance to structural materials operating under extreme thermomechanical conditions.

Core Competencies

Atomistic & Multiscale Modelling	⚙️	Molecular Dynamics (MD), Discrete Dislocation Dynamics (DDD), Atomistic Simulations, Microstructure Evolution, Defect Mechanics
Computational Modeling Tools	🔧	LAMMPS, OVITO, Atomsk, ABAQUS, ANSYS, COMSOL
Programming & Scientific Computing	📄	Python, High-Performance Computing (HPC, MPI/OpenMP), C++, Fortran, Bash, Parallel Simulation Frameworks, Git
Data Analysis & Visualization	📊	NumPy, Pandas, SciPy, Matplotlib, Automated Post-Processing, High-Volume Simulation Data Analysis
Tools & Environments	💻	Linux (Ubuntu), High-Performance Computing Clusters, Git/GitHub, LaTeX

Experience

April 2026 – July 2026	Visiting Researcher , Merrimack College, Department of Mechanical and Electrical Engineering, North Andover, MA, USA ➤ Conducted large-scale molecular dynamics simulations of temperature-dependent crack-tip deformation and ductile-to-brittle transition behavior in tungsten and tungsten–rhenium alloys. ➤ Developed automated LAMMPS workflows for high-performance computing clusters to examine the effects of temperature, alloy composition, and defect structure on mechanical response. ➤ Built custom Python pipelines using NumPy and Pandas for trajectory processing, stress–strain analysis, defect characterization, and visualization of large atomistic datasets. ➤ Investigated the roles of dislocation activity, crack-tip plasticity, and local structural evolution in controlling stress relaxation and fracture behavior. ➤ Complemented the atomistic analysis with a representative three-dimensional polycrystalline DDD framework for studying Frank–Read source activation, dislocation multiplication, and density-dependent hardening.
Jan 2025 – Dec 2025	Junior Research Fellow (DRDO Project) , IIT Bombay, Aerospace Engineering, Mumbai, India ➤ Conducted research on the constitutive modelling and high strain-rate response of Shape Memory Alloy (SMA)-based composites for aerospace and defense applications. ➤ Performed dynamic impact experiments using Split Hopkinson Pressure Bar (SHPB) systems to characterize material behaviour under high strain-rate loading. ➤ Integrated experimental observations with numerical simulations to calibrate and validate constitutive models for aerospace structural applications. ➤ Assisted in project execution, data analysis, and mentoring of junior researchers in experimental characterization and computational modelling.

- Jan 2019 – June 2025** **Ph.D. Research Scholar, IIT Bombay, Aerospace Engineering, Mumbai, India**
- Designed and executed large-scale atomistic simulations using **LAMMPS** on high-performance computing platforms to investigate generalized stacking fault energies, defect mobilities, and dislocation core structures.
 - Developed a physics-informed hierarchical multiscale framework that maps atomistic-derived parameters (drag coefficients, obstacle strengths, and lattice strain fields) directly into mesoscale continuum models.
 - Engineered automated post-processing and data-mining pipelines using **Python** to parse high-volume molecular dynamics (MD) trajectory data, track defect coordinates, and analyze structural evolution via OVITO.
 - Authored a custom parallel polycrystalline Discrete Dislocation Dynamics (DDD) code coupled with Finite Element Analysis (FEA) to simulate 3D defect interactions and slip localization near internal interfaces.
 - Utilized parallel computing (MPI) to scale atomistic simulations up to multi-million atom systems, establishing quantitative relationships between alloy microstructures and mechanical response under high stress fields.
- Jan 2019 – Dec 2023** **Graduate Teaching Assistant, IIT Bombay, Aerospace Engineering, Mumbai, India**
- Assisted instruction in Continuum Mechanics, Finite Element Methods (FEM), and Multiscale Modeling for undergraduate and postgraduate engineering courses.
 - Developed and supervised computational laboratory exercises involving numerical methods, finite element analysis, and scientific programming.
- June 2016 – Dec 2018** **Assistant Professor, CMR Institute of Technology, Mechanical Engineering, Bangalore, India**
- Taught undergraduate courses in Finite Element Methods, Experimental Stress Analysis, and Solid Mechanics.
 - Developed computational laboratory modules using ANSYS, MATLAB, and Python to support engineering analysis and numerical modelling.
- June 2014 – July 2016** **Graduate Teaching Assistant & M.Tech. Researcher, IIT Kharagpur, Mechanical Engineering, Kharagpur, India**
- Developed and validated computational models using ABAQUS and ANSYS for structural mechanics and vibration analyses.
 - Conducted laboratory instruction in Finite Element Modelling, Vibration Analysis, and Tribology, integrating theoretical concepts with numerical simulations.

Education

- Jan 2019 – Dec 2025**  Indian Institute of Technology - Bombay, Mumbai, India
 Doctorate, Aerospace Engineering
Thesis : Multiscale Modeling of Prior Particle Boundaries in Nickel-based Superalloys
CGPA : 9.36/10 *Funding* : MHRD Teaching Assistantship through Project
Key Subjects : Multiscale Modeling, Continuum Mechanics, Plasticity, Computational Mechanics
Advisors : Prof. Amuthan A. Ramabathiran, Prof. P. J. Guruprasad
- Jun 2014 – Jul 2016**  Indian Institute of Technology - Kharagpur, West Bengal, India
 M.Tech., Mechanical Engineering (Mechanical System Design)
Thesis : A Mesh-Free Model for Static Analysis of Smart Composite Beams
CGPA : 8.85/10 *Funding* : MHRD GATE Fellowship
Key Subjects : Solid Mechanics, Fracture Mechanics, Finite Element Methods, Composites, Elasticity, Vibration
Advisor : Prof. Manas Chandra Ray

Publications and Conference Contributions

- [1] **Mistry, D. A.** & Ramabathiran, A. A.,
Size-Dependent Power Laws for Edge Dislocations in Nickel Superalloys : A Molecular Dynamics Study,
Computational Materials Science, 259, 114122 (2025). DOI : 10.1016/j.commatsci.2025.114122.
- [2] **Mistry, D. A.**, Tak, T. N., Sankarasubramanian, R., & Guruprasad, P. J.,
A Discrete Dislocation Dynamics Study of Prior Particle Boundary Effects in Ni-based Superalloys,
Journal of Engineering Materials and Technology, pp. 1–55 (2026). Published online : August 4, 2026. DOI : 10.1115/1.4072448.
- [3] **Mistry, D. A.** & Mahata, A.,
Crack-Tip Plasticity and Ductile-to-Brittle Transition in W and W–Re Systems through Atomistic Fracture and Mesoscale Dislocation Dynamics.
Manuscript in Preparation, 2026.
- [4] **Mistry, D. A.**, Tak, T. N., & Guruprasad, P. J.,
Polycrystalline Discrete Dislocation Dynamics of Ni-Based Superalloys : Interactions with Prior Particle Boundaries and Second Phases Using Atomistically Informed Inputs.
Manuscript in Preparation.
- [5] **Mistry, D. A.**, Mahata, A., Ramabathiran, A. A., Guruprasad, P. J., & Tak, T. N.,
Multiscale Modeling of Dislocation–Precipitate–PPB Interactions in Powder-Metallurgy Nickel Superalloys.
Engineering Mechanics Institute Conference (EMI 2026), June 2–5, 2026, Boulder, Colorado, USA.
- [6] **Mistry, D. A.**, Tak, T. N., & Guruprasad, P. J.,
Hierarchical Multiscale Modeling of Plasticity in Ni-Based Superalloys : The Combined Role of Prior Particle Boundaries and Second Phases.
14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), October 12–16, 2025, IIT Roorkee, India.
- [7] **Mistry, D. A.**, Guruprasad, P. J., & Ramabathiran, A. A.,
An Atomistically Informed Discrete Dislocation Dynamics Study of Prior Particle Boundaries in Ni-Based Superalloys.
17th Conference on Computational Plasticity (COMPLAS), September 5–7, 2023, Barcelona, Spain.

Certifications

Jan 2020		Using Python for Research , edX, HarvardX, Harvard University.
Aug 2020		The Unix Workbench , Coursera, Johns Hopkins University.
Aug 2020		Material Behavior , Coursera, Georgia Institute of Technology.
May 2018		A Hands-on Introduction to Engineering Simulations , edX, CornellX, Cornell University.

Awards, Achievements, and Honorable Mentions

2025		Membership : Member of The Minerals, Metals & Materials Society (TMS), USA
2025		Institution of Eminence Funds, IIT Bombay : Awarded to attend the international conference <i>TMS 2025</i> , USA
2023		IIT Bombay Travel Grant : Awarded to attend the international conference <i>COMPLAS 2023</i> , Spain
2019		MHRD Teaching Assistantship Through Project : Awarded by the Industrial Research and Consultancy Centre (IRCC)
2014		All India Rank 69 : Achieved in the Graduate Aptitude Test in Engineering (GATE)
2013		Permanent Membership : Awarded permanent membership as an Associate Member at the Aero-nautical Society of India